



Seismic Detection and Characterisation of Strokkur Geyser Activity

Volcanoseismology Project Report

Student: Marwan Karam Saeed Saad

Under Supervision of Prof.DR. Eva Eibl

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University of Potsdam

Abstract

This project analysed continuous seismic data recorded around Strokkur geyser in Iceland in order to identify recurrent transient signals that may be associated with geyser eruptions. The dataset contained three-component recordings from stations GE2, GE4, and GE5 between 1 and 7 January 2026, sampled at 100 Hz. The processing followed the workflow introduced in the course: data inspection, demeaning, linear detrending, tapering, band-pass filtering, STA/LTA detection, coincidence verification, recurrence analysis, and particle-motion analysis.

A 10-40 Hz band-pass filter was selected after inspecting the waveform, spectrogram, and frequency spectrum of a representative transient. Event detection was performed using $STA = 2$ s, $LTA = 20$ s, trigger-on = 4, and trigger-off = 1.5. Requiring coincidence at all three stations produced 3341 coherent transients. These detections were grouped into 2904 seismic episodes, and a conservative two-minute separation criterion produced 1946 principal eruption candidates. The recurrence intervals of the principal candidates were mainly concentrated between approximately 3 and 6 minutes, with a mean of 4.78 minutes and a median of 4.27 minutes.

A strong event recorded at GE2 on 2 January 2026 at 09:50:23.67 UTC was selected for detailed analysis. Moving-window horizontal particle motion showed a highly rectilinear interval between 0.10 and 0.40 s after the detected onset. The rectilinearity was 0.909 and the dominant horizontal polarization axis was $344.5^\circ/164.5^\circ$. Because the available StationXML file contained only station coordinates and not an independent coordinate for the Strokkur vent, the 180° directional ambiguity could not be resolved.

1. Introduction

Geyser eruptions are repeating hydrothermal processes that can generate clear seismic signals. These signals may be caused by fluid movement, pressure changes, bubble collapse, and the eruption itself. Because the activity repeats over time, continuous seismic monitoring can be used to detect individual events and to study their recurrence behaviour.

The main aim of this project was to build a complete and reproducible seismic workflow for the Strokkur dataset. The analysis focused on four main questions: whether coherent transient signals could be detected at several stations, how often the signals occurred, what the waveform and frequency characteristics of a representative event looked like, and whether the horizontal particle motion showed a stable dominant orientation.

The report is based directly on the final Python/ObsPy notebook. The intention was not to apply the methods learned in the course in a clear and scientifically structured way.

1.1 Objectives

- Inspect the available stations, channels, sampling rates, and recording durations.
- Prepare the seismic data using demeaning, linear detrending, tapering, and band-pass filtering.
- Detect transient signals using STA/LTA triggering.
- Verify detections using three-station coincidence.
- Group related detections and construct an eruption-candidate catalogue.
- Analyse event rates and recurrence intervals.
- Study the waveform, spectrogram, and particle motion of one representative event.
- Use StationXML metadata to confirm station coordinates and component orientations.

2. Data

The dataset consisted of MiniSEED files from the temporary network code 2Z. Three seismic stations were available: GE2, GE4, and GE5. Each station recorded the east-west component (HHE), north-south component (HHN), and vertical component (HHZ). The sampling rate was 100 Hz for all traces.

The recordings covered 1-7 January 2026. The first six days were almost complete 24-hour records, while the final day was shorter and had slightly different end times at the three stations. In total, 63 traces were loaded, corresponding to 3 stations $\times$ 3 components $\times$ 7 days.

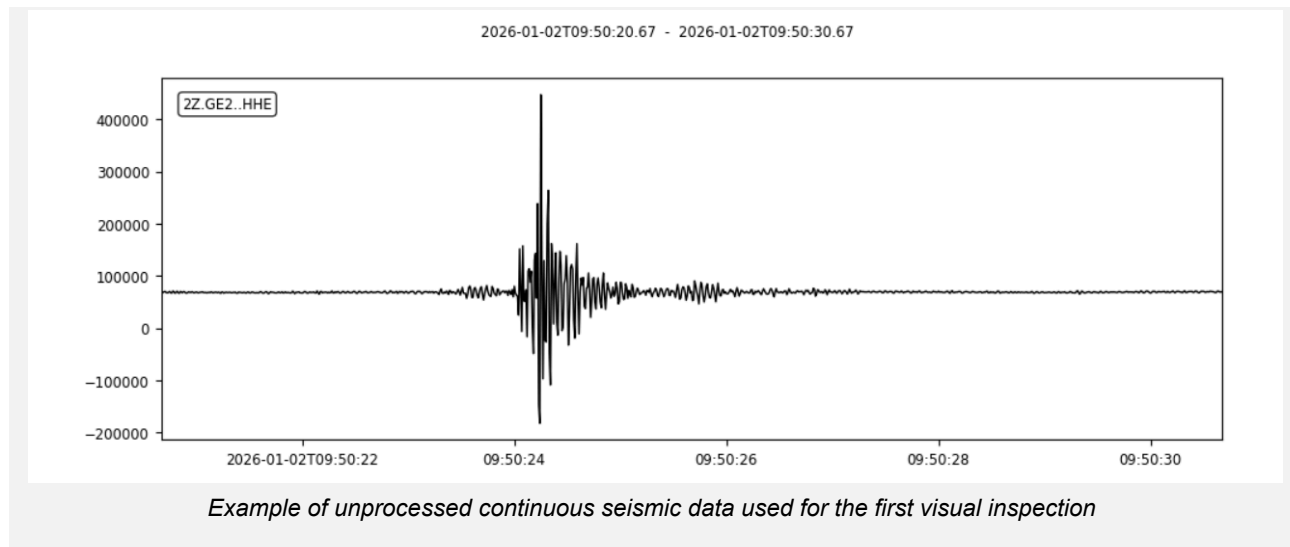
Station	Latitude	Longitude	Elevation (m)	Components
GE2	64.312970	-20.301258	143.8	HHE, HHN, HHZ
GE4	64.312499	-20.300098	117.1	HHE, HHN, HHZ
GE5	64.312783	-20.302133	125.9	HHE, HHN, HHZ

The StationXML metadata confirmed that HHN is oriented toward geographic north (0°), HHE toward east (90°), and HHZ is vertical. This was important for interpreting the horizontal particle-motion result.

3. Methodology

3.1 Data Loading and Initial Inspection

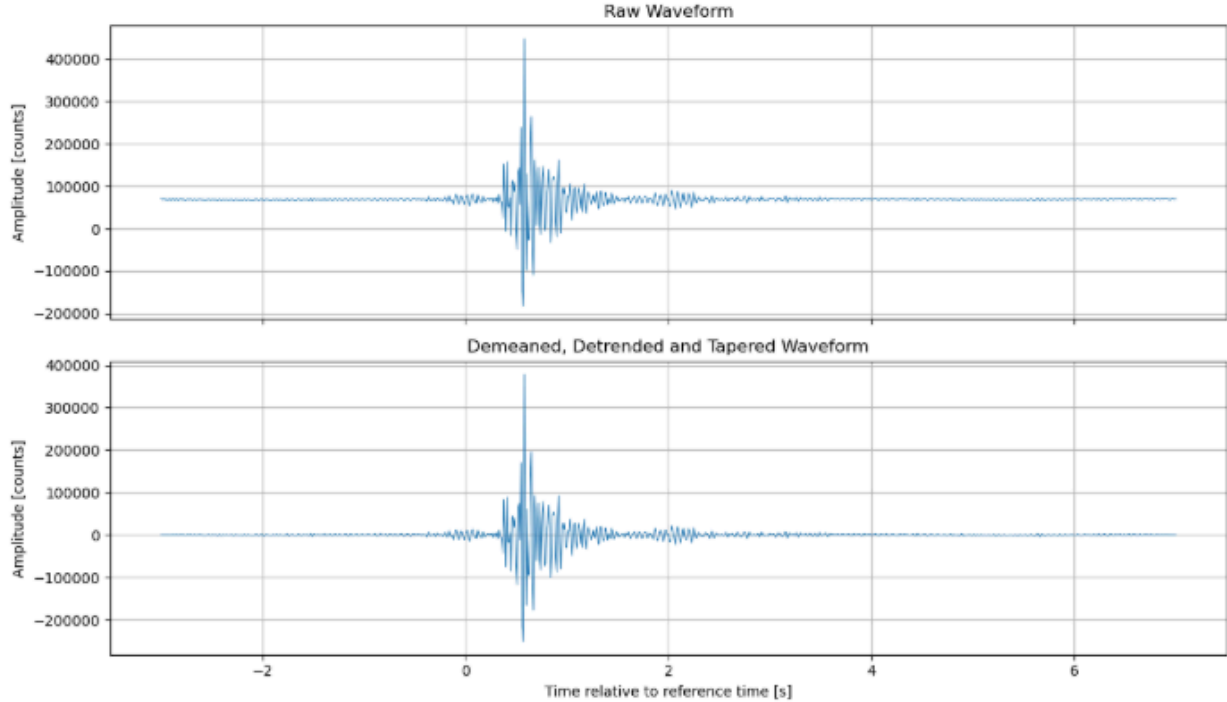
All seismic files in the project folder were read with ObsPy and combined into one stream. The stream was inspected to confirm the station names, channel names, sampling rates, start times, end times, and number of samples. This first step was necessary because the final day was incomplete and therefore the event rates had to be normalised by the actual recording duration.



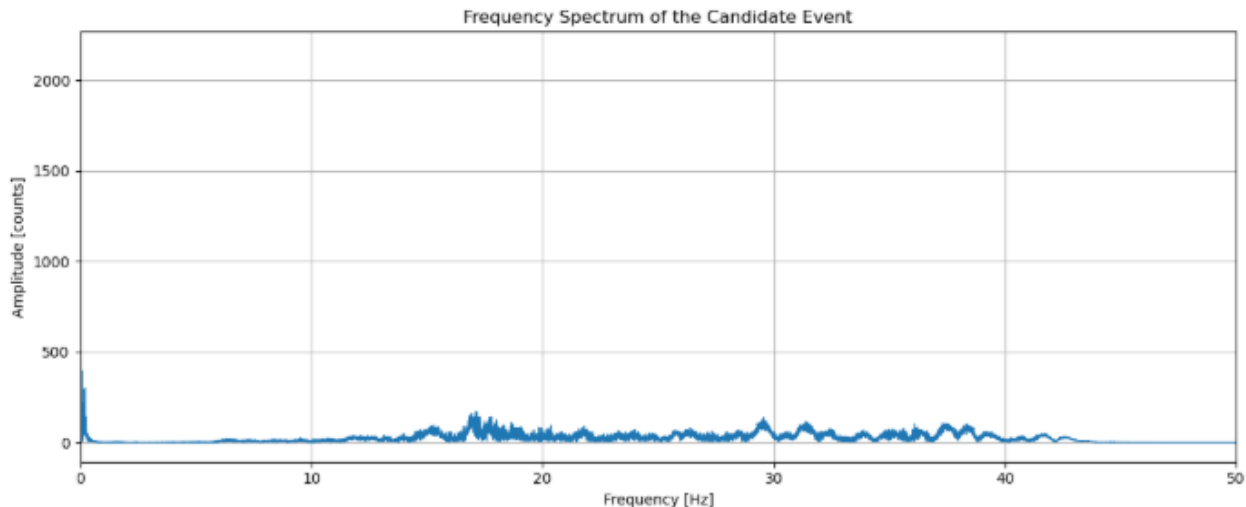
3.2 Preprocessing

Preprocessing was performed on copies of the original traces so that the raw recordings remained unchanged. Each selected trace was demeaned to remove its constant offset and linearly detrended to remove slow drift. A Hann taper was then applied to reduce edge effects before filtering.

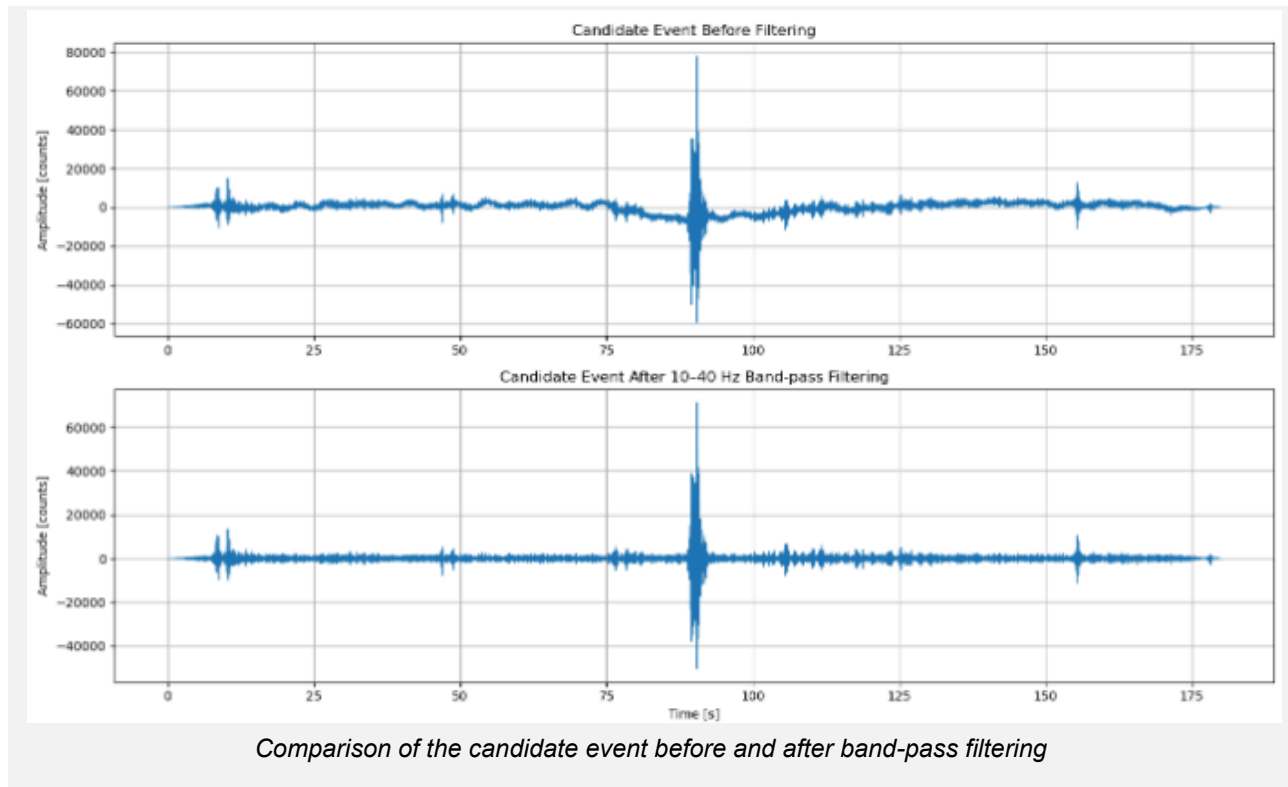
The frequency content of a clear transient was inspected in both the time and frequency domains. The event showed useful energy mainly within the higher-frequency part of the available band. Based on this inspection, a 10-40 Hz band-pass filter was applied with four corners and zero-phase filtering. The zero-phase option avoided a phase shift in the waveform.



Effect of the preprocessing steps on a representative transient.



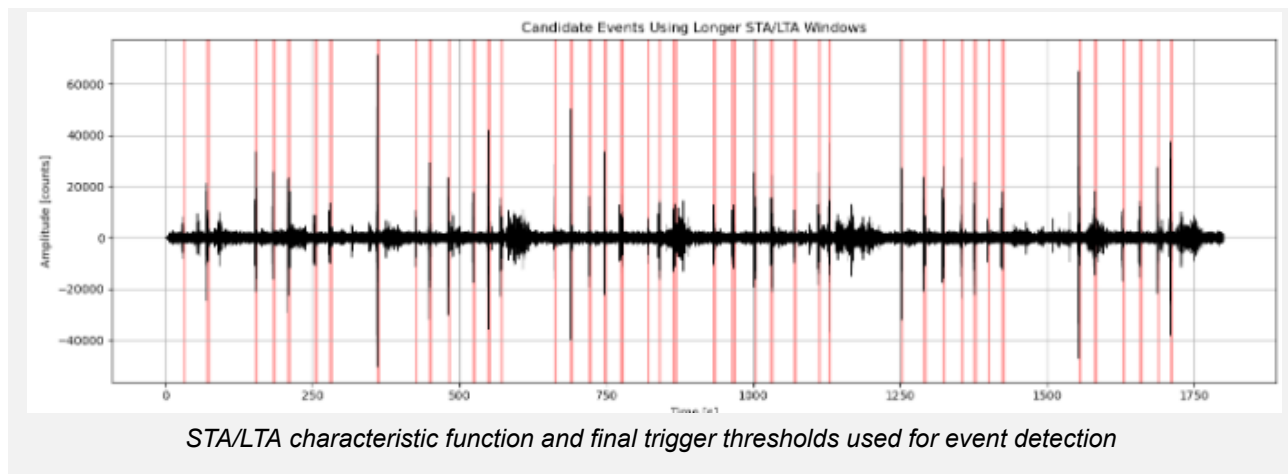
Frequency spectrum used to support the selection of the 10-40 Hz band-pass filter



3.3 STA/LTA Event Detection

Transient detection was performed using the short-term average/long-term average method. STA/LTA compares the recent signal level with a longer background level and is therefore suitable for detecting sudden increases in seismic amplitude. The final parameters were STA = 2.0 s, LTA = 20.0 s, trigger-on = 4, and trigger-off = 1.5.

The parameters were tested first on shorter waveform segments before being applied to the complete dataset. The longer STA and LTA windows were chosen to reduce the influence of very short noise spikes and to identify the more sustained transient signals seen in the data.



3.4 Three-Station Coincidence Detection

A detection on one channel alone may be caused by local noise. To increase confidence, the filtered HHE components from GE2, GE4, and GE5 were processed together using coincidence triggering. A transient was accepted only when the coincidence sum reached three, meaning that it was recorded at all three stations.

This step produced 3341 coherent three-station transients over the full recording period. The daily numbers were 398, 460, 571, 562, 576, 566, and 208 detections for days 1-7 respectively. The lower number on day 7 was expected because the recording was shorter.

3.5 Grouping of Related Detections

One physical eruption can produce more than one nearby seismic trigger. Therefore, triggers separated by less than 30 s were grouped into one seismic episode. This reduced the 3341 coherent transients to 2904 grouped episodes.

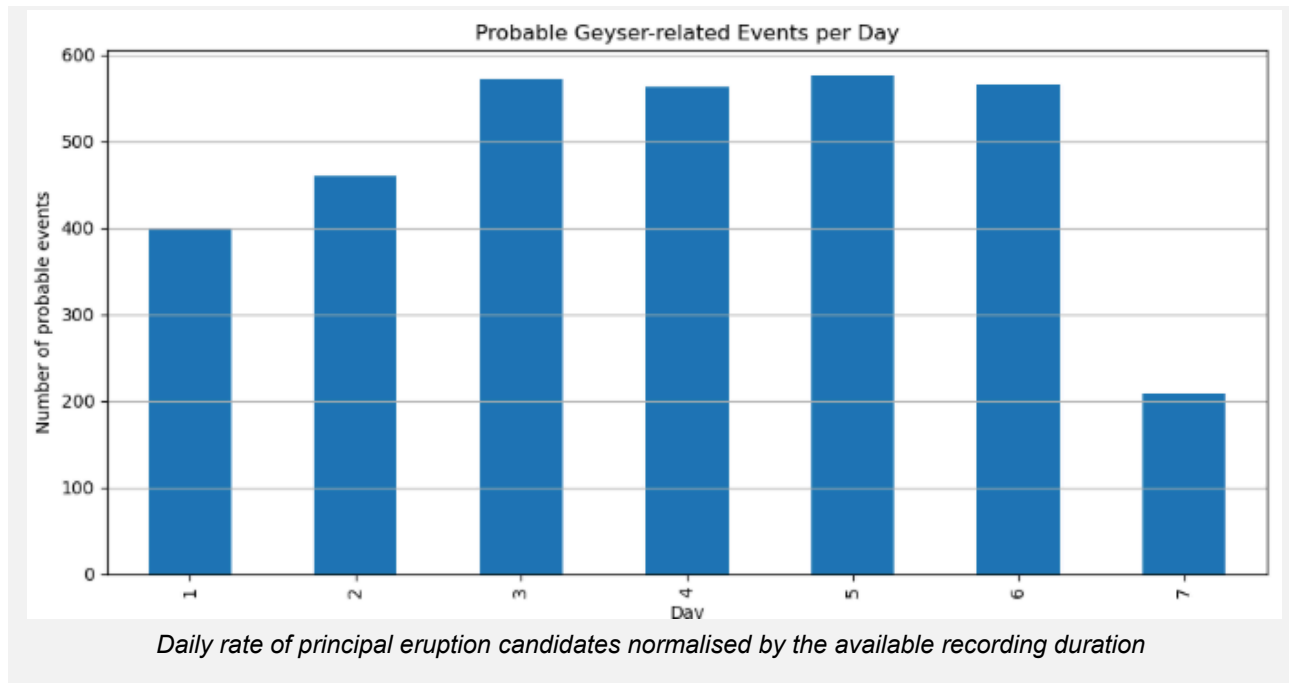
A second conservative selection was then applied. Episodes occurring within 120 s were treated as one cluster, and the episode with the highest mean network peak amplitude was retained. This produced 1946 principal eruption candidates. The result was described as conservative because the two-minute rule may remove some genuine eruptions that occurred close together. For this reason, the report keeps both the grouped-episode result and the principal-candidate result.

4. Results

4.1 Detection Catalogue

The complete workflow reduced the data in three stages. First, 3341 coherent transients were detected at all three stations. Second, these were grouped into 2904 seismic episodes. Third, the conservative separation rule produced 1946 principal eruption candidates.

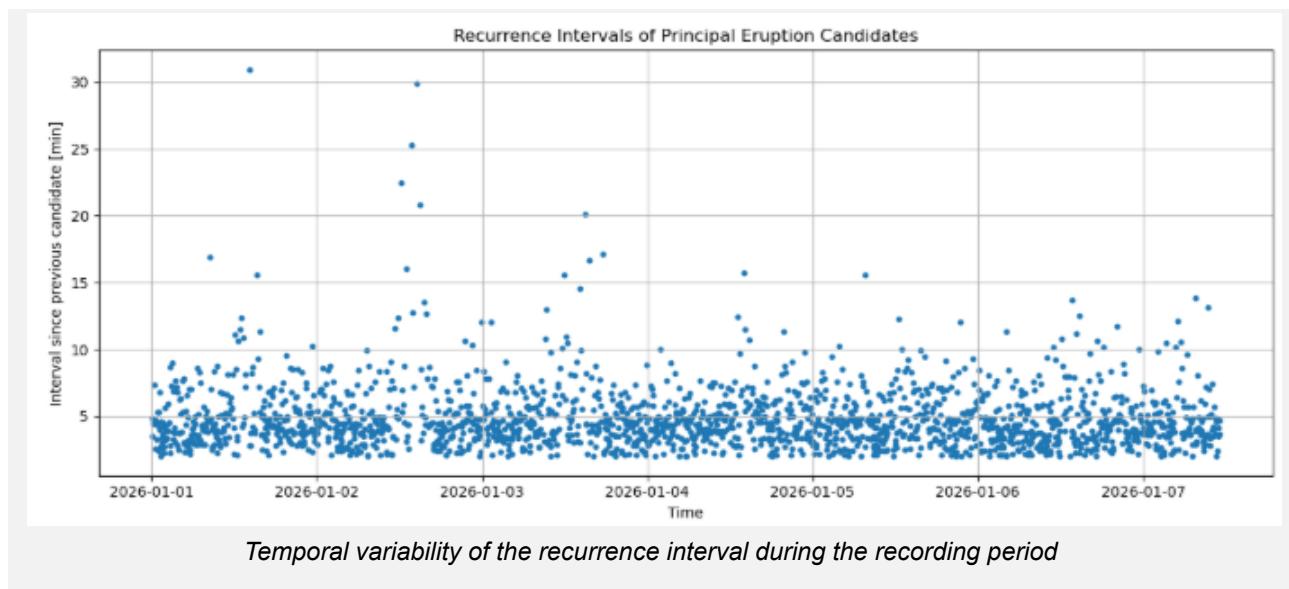
Processing stage	Number of events	Interpretation
Three-station coincidence transients	3341	Coherent triggers recorded at GE2, GE4, and GE5
Grouped seismic episodes	2904	Nearby detections merged within 30 s
Principal eruption candidates	1946	Strongest episode retained within each 120 s cluster



4.2 Recurrence Intervals

The recurrence interval was calculated from the time difference between consecutive principal eruption candidates. The mean recurrence interval was 4.78 minutes and the median was 4.27 minutes. Most intervals were concentrated between approximately 3 and 6 minutes.

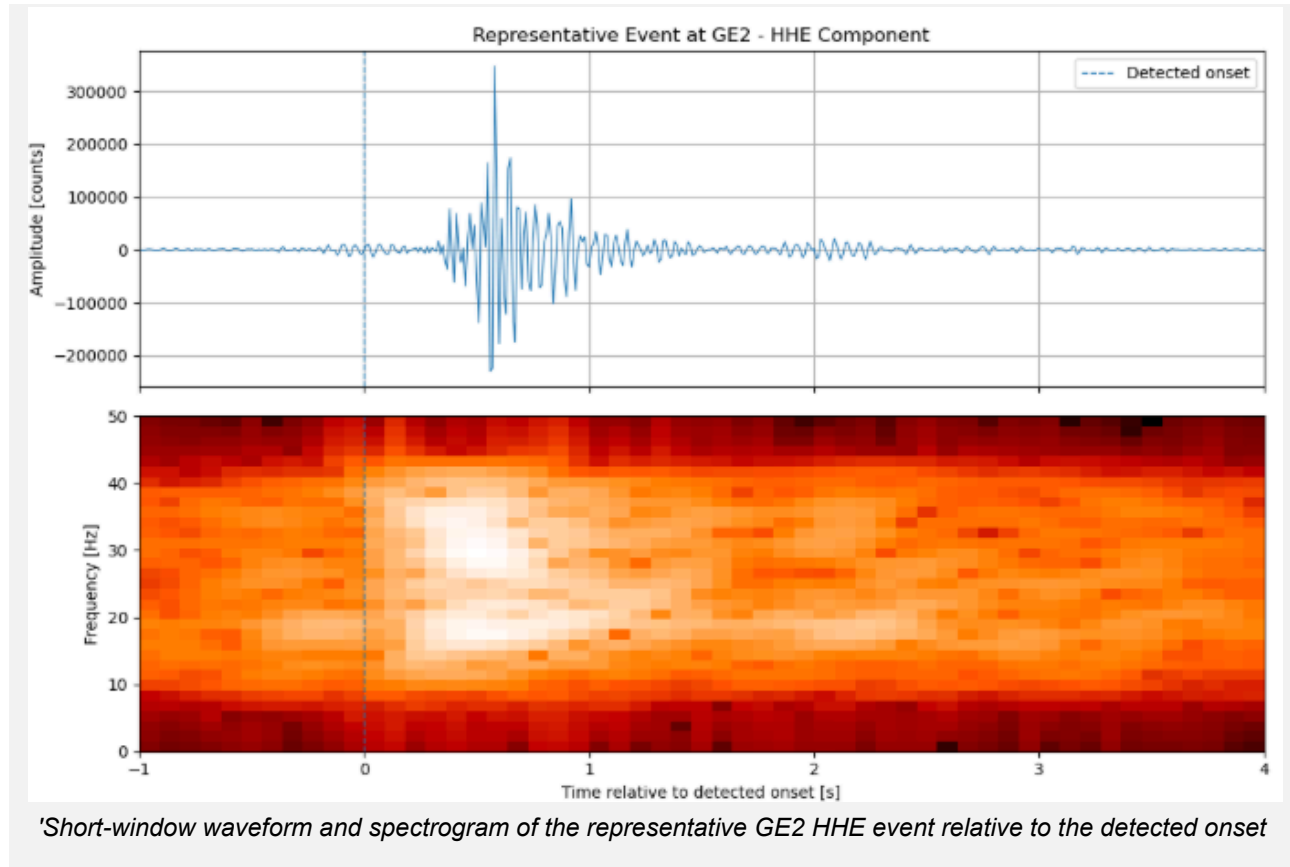
The mean was slightly larger than the median, which indicates that some longer quiet intervals increased the average. Short intervals were also present, but many of the very closely spaced signals had already been reduced by the grouping and conservative selection steps.

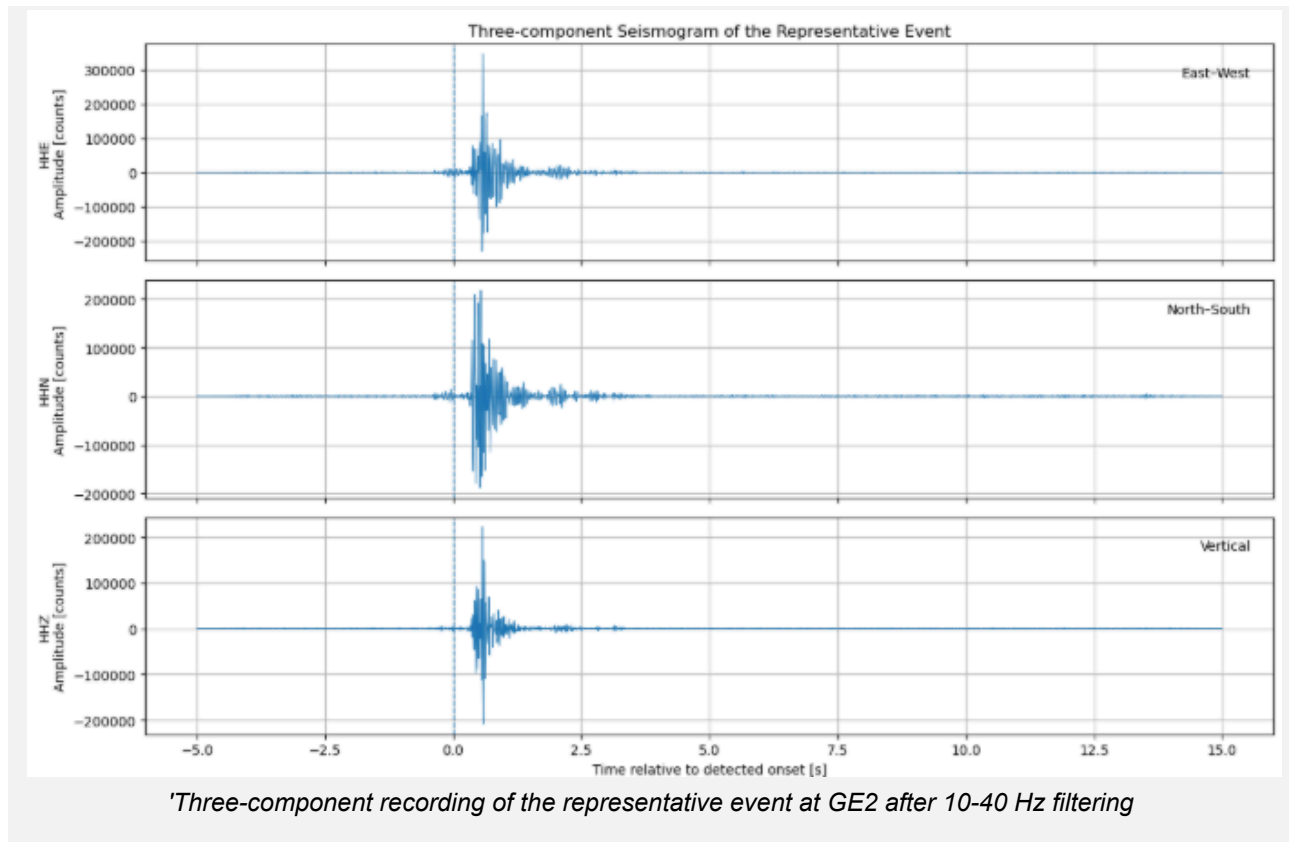


4.3 Representative Event

A strong and isolated event recorded on 2 January 2026 at 09:50:23.67 UTC was selected for detailed analysis. The event was clearly visible at all three stations and had a distinct onset followed by a decaying coda. Station GE2 was used for the detailed three-component and particle-motion analysis.

The final waveform and spectrogram plot was shortened to the interval from 1 s before to 4 s after the detected onset. This shorter window made the onset and the main part of the signal easier to interpret. The highest-amplitude motion occurred shortly after the automatic trigger time, and the spectrogram showed energy over a broad part of the selected 10-40 Hz band.



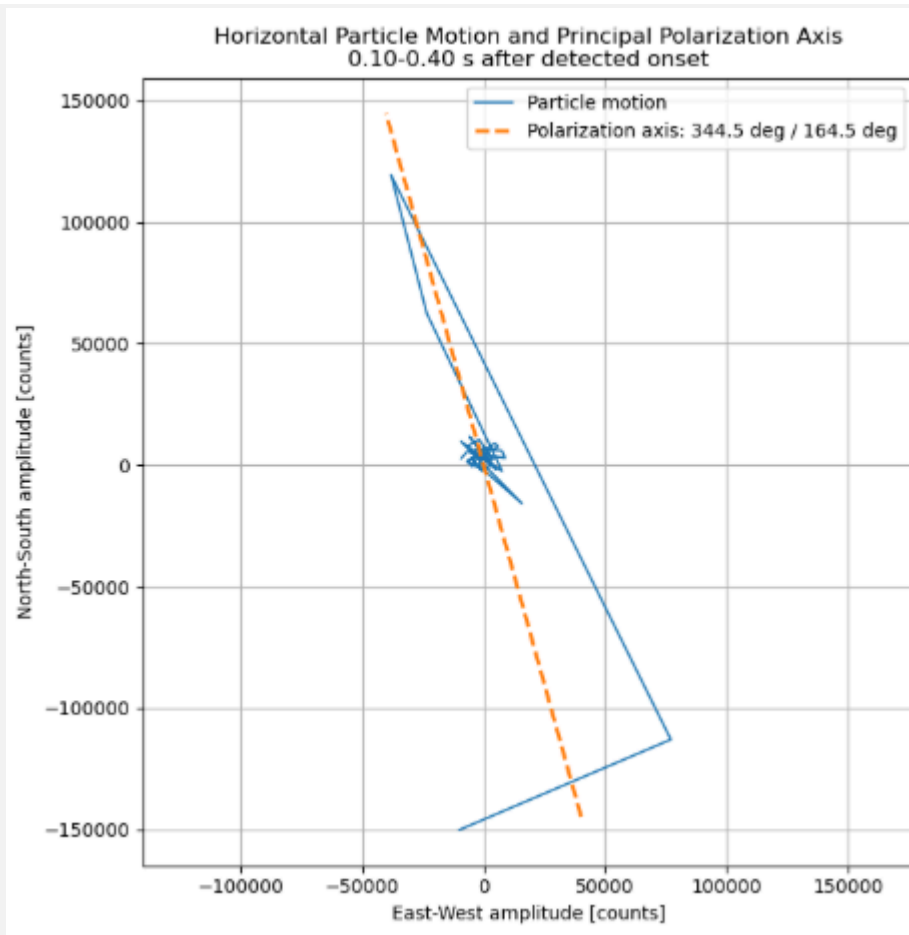


4.4 Horizontal Particle Motion and Polarization

The horizontal particle motion was calculated from the east-west and north-south components at GE2. A moving 0.30 s window with a 0.05 s step was scanned through the first 2 s after the detected onset. Principal-component analysis was applied in each window, and rectilinearity was calculated from the ratio between the two horizontal eigenvalues.

The best result occurred between 0.10 and 0.40 s after onset. The rectilinearity was 0.909, showing that the horizontal motion was strongly concentrated along one axis during this short interval. The corresponding horizontal orientation was 344.5° or the opposite direction, 164.5° .

Particle-motion analysis defines an axis and therefore has an inherent 180° ambiguity. The supplied StationXML file confirmed that HHE and HHN were geographically oriented, but it did not provide an independent coordinate for the Strokkur vent. As a result, it was not possible to decide which end of the axis pointed toward the source.



Horizontal particle motion at GE2 between 0.10 and 0.40 s after onset, showing the 344.5°/164.5° polarization axis

5. Discussion

The three-station coincidence requirement was one of the most important parts of the workflow because it reduced the chance that local station noise would be interpreted as geyser activity. The large number of coherent detections shows that recurrent seismic signals were recorded throughout the week.

The difference between 3341 transients, 2904 episodes, and 1946 principal candidates demonstrates that the event count strongly depends on how nearby detections are grouped. The 30 s grouping mainly removed multiple triggers related to the same episode. The 120 s selection was more conservative and may also have removed real closely spaced eruptions. Therefore, the principal catalogue should be interpreted as a lower, cleaner estimate rather than the only possible event count.

The recurrence distribution was physically reasonable for a regularly repeating geyser signal because most intervals were concentrated within a limited range rather than being completely random. However, the spread and occasional long intervals show that the system was not perfectly periodic.

The representative event showed a clear transient waveform and broad high-frequency energy after filtering. The strong rectilinearity in the short 0.10-0.40 s window suggests that a stable horizontal motion was present immediately after onset. At the same time, this result should not be extended to the complete event because the later coda showed more complicated motion.

A unique back azimuth was not claimed. The measured value is a polarization axis, not a fully resolved source direction. This cautious interpretation is important because the vent coordinate was not included in the supplied metadata.

6. Limitations

- The final day was shorter than the other recording days, so raw daily counts cannot be compared without normalising by recording duration.
- STA/LTA parameters were selected after practical testing and visual inspection, so a different parameter set could produce a different number of detections.
- The 120 s principal-candidate criterion is conservative and may remove genuine closely spaced eruptions.
- Only one representative event was studied in detail, so the polarization result does not describe every event in the catalogue.
- The horizontal polarization has a 180° ambiguity, and the Strokkur vent coordinate was not available in the StationXML file.
- No independent eruption log or visual observation record was available for validating every seismic candidate.

7. Conclusion

This project successfully applied the main volcanoseismology methods from the course to one week of continuous Strokkur recordings. The data were inspected, preprocessed, filtered, and analysed using STA/LTA and three-station coincidence detection. A total of 3341 coherent transients were detected, which were reduced to 2904 grouped seismic episodes and 1946 conservative principal eruption candidates.

The principal candidates had a mean recurrence interval of 4.78 minutes and a median of 4.27 minutes, with most intervals between approximately 3 and 6 minutes. A representative event recorded on 2 January 2026 showed a clear onset, strong energy within the 10-40 Hz band, and a strongly rectilinear horizontal particle-motion window.

The dominant horizontal polarization axis was 344.5°/164.5°, but the sourceward direction could not be resolved because an independent vent coordinate was not available. Overall, the analysis produced a coherent event catalogue and a scientifically cautious characterisation of the recurring seismic activity recorded around Strokkur.

Appendix A. Main Processing Parameters

Parameter	Value
Sampling rate	100 Hz
Preprocessing	Demean, linear detrend, Hann taper
Band-pass filter	10-40 Hz, 4 corners, zero phase
STA window	2.0 s
LTA window	20.0 s
Trigger-on threshold	4.0
Trigger-off threshold	1.5
Coincidence requirement	3 stations
Episode grouping	30 s
Principal-candidate separation	120 s
Polarization moving window	0.30 s
Polarization step	0.05 s